

Sheel Patel

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SUMMARY

Master's graduate in Computer Science with experience designing and deploying full-stack, cloud-native applications and scalable backend systems. Skilled in Python, C++, and distributed systems, with hands-on expertise in AWS/GCP, containerized deployments, and API-driven architectures. Background in machine learning and data engineering enables building intelligent, production-ready systems.

EDUCATION

Master of Science in Computer Science, *New York University*

May 2025

Relevant Coursework: Machine Learning, Deep Learning, Database Systems, Big Data, Cloud Computing, Applied ML for Finance.

B.Tech in Computer Science and Engineering, *SRM Institute of Science and Technology*

May 2023

TECHNICAL SKILLS

Programming — Python, C++, SQL

Backend & Cloud — AWS (S3, Lambda, DynamoDB, EC2, API Gateway, Bedrock), Google Cloud, Docker, Kubernetes, Heroku, Flask, FastAPI

Frontend & Tools — HTML, CSS, Streamlit, Git, GitHub, Jupyter, Linux (Ubuntu)

Databases — PostgreSQL, MongoDB, Redis

Machine Learning and Data — PyTorch, TensorFlow, Keras, Scikit-learn, Hugging Face, LangChain, OpenCV, PySpark, Spark

EXPERIENCE

Graduate Research Assistant, *New York University*

Nov 2024 – May 2025

- Built a **modular time-series forecasting system** inspired by ARIMAX, engineering components (AR, I, MA, X) as standalone services with neural network backends.
- Conducted benchmarking pipelines across multiple datasets, optimizing system performance by **10–15%** compared to statistical and deep learning baselines.

Artificial Intelligence Research Intern, *SRM Institute of Science and Technology*

Jul 2022 – Aug 2023

- Engineered a **robust data pipeline** for medical imaging, integrating preprocessing, augmentation, and hyperparameter tuning.
- Deployed CNN models achieving **98%** F1-score on **500+** X-ray images, improving diagnostic classification.

Cloud & ML Engineering Intern, *Wastefull Insights*

Sep 2022 – May 2023

- Designed and deployed a **real-time object detection system** (>15 FPS) on AWS EC2/S3, enabling scalable waste classification.
- Improved system robustness by **10%** via iterative engineering and feedback integration.
- Analyzed **10K+** image segments to identify error cases, guiding next-generation data strategy.

PROJECTS

Research Paper Summarizer

- Developed an end-to-end platform with AWS backend (S3, Lambda, DynamoDB, API Gateway, Bedrock) for research paper summarization and search.
- Implemented REST APIs for LLM-driven summarization, keyword extraction, and search with OpenSearch.

Image and Audio Caption Generation

- Built multimodal application with image-to-text captioning and text-to-speech conversion.
- Deployed backend as REST API (Flask) and integrated with frontend web app, demonstrating full-stack ML productization.

Loan Default Prediction

- Built a scalable ML pipeline for financial risk analysis on **1M+** records.
- Engineered **20+** financial features and deployed models (logistic regression, Gradient Boost) with Pandas/Scikit-learn pipelines.

LEADERSHIP

Data Science Community SRM, *President*

Jul 2021 – Jul 2022

- Directed a 100+ member tech community, fostering collaboration across academic and industry-focused projects.
- Organized 10+ workshops, hackathons, and ML research events, driving peer learning and exposure to applied AI.
- Mentored students on projects and research, improving the technical depth and research output of community members.

COURSES UNDERTAKEN

Algorithmic Toolbox, *UC San Diego*

Python for Everybody, *UMich Ann Arbor*

Machine Learning, *Stanford University*

Deep Learning, *DeepLearning.AI*